

Danmaku Avatar: Enabling Asynchronous Co-viewing Experiences in Virtual Reality via Danmaku

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Abstract

As social interaction needs evolve, danmaku comments on 2D screens have gained popularity in Asia, offering users an asynchronous virtual co-viewing experience. With the advancement of virtual reality (VR) technology, the demand for similar co-viewing experiences in VR environments has grown. In this study, we developed the Danmaku Avatar system to adapt danmaku for VR settings, enhancing asynchronous shared viewing experiences. Comparative experiments revealed that danmaku avatars significantly increased social presence and fostered a greater desire for communication. These findings provide a foundation for further refining danmaku representations in VR, advancing virtual social interactions and immersive viewing experiences.

CCS Concepts

• **Human-centered computing** → *Collaborative and social computing; Interaction design.*

Keywords

Virtual Reality, Asynchronous Social Interaction, Asynchronous Co-viewing, Danmaku

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1 Introduction

Traditionally, co-viewing with family or friends has been a fundamental aspect of the television (TV) and video viewing experience [25]. This form of co-viewing typically involves real-time interaction and communication, such as watching TV together in a living room or using tools like Teleparty¹ for synchronized remote viewing [24]. However, modern life's busyness and varied schedules often make it challenging to coordinate synchronous viewing sessions [28]. Asynchronous co-viewing [29] has emerged as a viable alternative, enabling individuals to watch the same content at different times while still engaging in shared experiences and interactions. Platforms such as YouTube² and Bilibili³ facilitate this through features such as comments, danmaku, and other social tools. Nonetheless, these systems typically operate on two-dimensional (2D) screens and rely on text-based interactions, which offer limited social presence [32] and may struggle to convey complex emotions and reactions [30]. Consequently, this mode of social engagement often lacks real-time responsiveness and depth.

With the continuous advancement of technology, immersive television viewing environments has gained significant popularity, offering users enhanced interactivity and a richer experience compared to text-based asynchronous co-viewing on 2D screens [42]. Asynchronous social virtual reality (VR) emerges as a promising solution, combining the convenience of asynchronous communication with the immersive presence of VR [39]. This concept has been explored in prior studies, which propose transforming or reconstructing recorded data into immersive environments to enable retrospective interaction and engagement [38, 39]. One notable example of asynchronous co-viewing is danmaku, a widely used commentary feature on Asian online video platforms [5, 14] (see Figure 2(a)). Danmaku comments are overlaid on video screens and

¹<https://www.teleparty.com/>

²<https://www.youtube.com/>

³<https://www.bilibili.com/>

synchronized with playback, allowing viewers to see remarks from previous audiences at corresponding moments. This functionality creates a pseudo-synchronous co-viewing experience, fostering a sense of shared engagement [4, 19, 21, 22, 40]. Recognizing the potential of danmaku to deliver shared viewing experiences, we aimed to adapt this feature into VR environments to facilitate asynchronous co-viewing of videos. However, transitioning text-based danmaku into an immersive VR setting poses a significant challenge, requiring innovative approaches to preserve and enhance its interactive and engaging nature.

Our work focuses on leveraging danmaku in VR scenarios to facilitate asynchronous co-viewing, targeting ordinary users seeking an improved co-viewing experience. To achieve this, we address the following key research questions: **RQ1**: What challenges do users encounter when co-viewing with danmaku on 2D screens? **RQ2**: What are users' expectations for danmaku-based co-viewing in VR environments? **RQ3**: Does the integration of danmaku into VR environments enhance the overall user experience? **RQ4**: How does the introduction of danmaku in VR influence viewer behavior?

To address **RQ1** and **RQ2**, we conducted an online survey to identify common challenges users face when co-viewing content with danmaku on 2D screens, as well as their expectations for watching video content in VR with danmaku. Our findings revealed that respondents view danmaku as a form of virtual co-viewer rather than just text comments. However, the text-only format and delayed responses diminished the sense of social presence. Respondents expressed a strong preference for a more immersive co-viewing experience in VR, particularly one with real-time interactive capabilities. Based on the results of our user survey, we invited 13 students (7 males and 6 females) from a local research university, all with experience in VR system design and familiarity with danmaku culture, to participate in discussions about the system design. Following these discussions, we voted for the *virtual agent with voice* design from the five best candidates (Figure 1). To address **RQ3** and **RQ4**, we designed and conducted a comparative experiment. The results indicated that danmaku avatars significantly enhanced social presence and increased users' desire for communication. Based on our research, we provide insights and guidelines for future researchers to develop a groundwork for refining danmaku representations in VR.

In summary, the key contributions of this study are as follows: (a) We conducted a formative study to explore users' needs and expectations for danmaku interaction and immersive asynchronous co-viewing experiences in VR. (b) We designed and implemented the Danmaku Avatar system, a VR-based asynchronous co-viewing platform that integrates danmaku data to enhance user engagement. (c) We designed and conducted a comparative experiment to evaluate the impact of danmaku avatars on the co-viewing experience.

2 Related Work

2.1 Asynchronous Co-Viewing

Asynchronous technology enables co-viewing in an asynchronous context [8, 38, 39]. Platforms like Twitter, YouTube, and other social media facilitate asynchronous co-viewing by allowing users to engage with videos through hashtags, retweets, and comments, thereby creating a shared viewing experience despite watching the

content at different times [29]. Asynchronous social interaction merges the flexibility of asynchronous communication with the immersive presence of VR [37–39]. Wang et al. [38, 39] explored the design space for asynchronous social interactions, particularly how modifying recorded group interactions can help integrate new users. Yan et al. [41] incorporated audience voice comments into VR cinema, demonstrating that voice comments from previous viewers can create an asynchronous co-viewing experience for current viewers. Several studies have examined the presentation of danmaku comments on 360° VR video screens [18] and the use of danmaku from online study videos to create virtual classmates [20]. However, these works have not systematically explored the broader impact of danmaku-facilitated co-viewing experiences, nor have they fully harnessed the immersive potential of VR to transform danmaku into a spatially simulated and interactive experience.

Users often struggle with synchronous co-viewing due to the challenge of coordinating schedules across different time zones and individual daily routines [9]. Co-viewing on mobile applications, such as phones and computers, usually leads to limited communication, with all interactions confined to the screen. Although previous studies have enriched social interactions in asynchronous social VR, few have focused on enhancing asynchronous co-viewing experiences. Our research addresses this gap by transforming recorded video comments, providing a more interactive and emotionally engaging asynchronous co-viewing experience.

2.2 Virtual Agents and Emotional Body Expression

Designing virtual agents, or embodied conversational agents (ECAs) with social characteristics, is a crucial focus in the human-computer interaction community, as most virtual or robotic agent applications need to exhibit some degree of social capability. The design process of virtual agents typically includes both verbal and non-verbal behaviors, including facial expressions, head movements, and posture. Defining these features generally requires establishing a correspondence between the verbal and non-verbal characteristics of virtual agents through emotional modeling [10, 34].

Bhattacharya et al. [3] proposed a method for translating text into emotive body gestures by extracting emotional features from the text, which in turn generates emotional gestures for virtual agents. Alanazi et al. [1] employed deep learning techniques, incorporating multimodal emotional cues to develop an embedded method for analyzing the empathetic behavior of virtual agents in socially dynamic contexts. Similarly, Ng et al. [26] proposed a new framework for generating full-body realistic avatars, capable of producing diverse facial expressions, body movements, and hand gestures based on voice input. Many existing studies focus on building emotion-body expression databases that correlate specific emotions with corresponding body movements [10, 35]. For instance, the Emilya Database [10] records 10,001 actions spanning 7 types of daily activities and 8 emotional expressions. Regarding the perception of virtual audience behavior, literature has proposed a valence-arousal emotional model to shape virtual audience behaviors [16], which can be used to describe perceived attitudes and guide the design of virtual audiences. In line with this research, Chollet et al. [6] offered a detailed model illustrating how nonverbal

behaviors are perceived and associated with emotional dimensions based on user perceptions. Glémarec et al. [12] utilized the valence-arousal model to design non-verbal behaviors for virtual audiences, representing specific audience attitudes such as indifference or enthusiasm. Drawing on these studies, we employed the valence-arousal model to inform the design of non-verbal behaviors for the danmaku avatar.

3 Formative Study

To address **RQ1** and **RQ2**, we conducted a formative study to understand users' needs and identify opportunities to enhance their asynchronous co-viewing experiences. After receiving approval from the Institutional Review Board (IRB), we began with an online survey in subsection 3.1 to explore users' needs and expectations. Building on this, subsection 3.2 through a workshop to discuss the system design.

3.1 Online Survey

We conducted an online survey to identify common challenges users face when co-viewing content with danmaku on 2D screens, as well as their expectations for watching video content in VR with danmaku. We received 132 valid responses, with an effective recovery rate of 78.4%. Among the respondents, 91 were female (68.94%) and 41 were male (31.06%), with ages ranging from 17 to 83 (mean age = 29.33). The participants were recruited through word of mouth, email, and social media, representing diverse social backgrounds and life experiences. The survey focused on three primary areas: asynchronous co-viewing needs, limitations of danmaku, and expectations of asynchronous co-viewing in an immersive environment. At the beginning of the questionnaire, we provided an introduction to the background and purpose of the survey.

Our findings revealed that current asynchronous co-viewing applications do not fully meet users' needs for real-time interaction and immersive. Although some platforms provide comment and discussion sections, these features often struggle to facilitate sustained conversations, lacking real-time interaction and emotional expression. While danmaku offers a form of asynchronous co-viewing, it has several limitations. In this survey, 67% of respondents had experience using danmaku. The survey revealed that users' motivations for using danmaku stem from various viewing needs: gain additional information and explanations (68.42%), enhance interaction and enjoyment (66.52%), share thoughts (47.37%), seek emotional support (42.11%), and understand the popular culture (26.32%). However, the current support danmaku systems have several limitations including danmaku obstructing the screen (63.16%), disruptions from repetitive comments (45.36%), overwhelming or uncategorized content leading to unwanted information (36.84%), and the standardized filtering mechanism limits user choices, preventing them from customizing their desired asynchronous viewing experience according to their preferences (35.24%). The majority of respondents (67.37%) have high expectations for asynchronous co-viewing experiences in an immersive environment. They generally believe that current 2D screen asynchronous co-viewing applications lack a true sense of social presence, which may lead to feelings of isolation or reduced engagement during the viewing experience.

3.2 Design Workshop

Based on our user survey results, we invited 13 students from a local research university to participate in discussions on system design. The group included 7 females and 6 males (mean age = 23.13, SD = 1.82) and was recruited through word of mouth and campus chat group. All participants had experience in VR system design and were familiar with danmaku culture.

During the first 30 minutes of the workshop, we first introduced participants with an overview of the research background and objectives, emphasizing that the goal was to generate several feasible design candidates. We also presented current research and practical cases related to VR and danmaku, as discussed in subsection 2.1, highlighting their strengths and weaknesses in inspiring creativity. Participants were then divided into groups of three to four members and given 40 minutes to brainstorm and propose at least two potential design solutions based on the three presentation methods: text, voice, and 3D objects. After each group presented their ideas, participants voted for their preferred solutions. Based on the votes, five candidate designs were selected: *scrolling danmaku*, *voice danmaku*, *danmaku bubbles*, *virtual agents with bubbles*, and *virtual agents with voice*. For the *scrolling danmaku*, they drew inspiration from live-streaming scenarios, displaying the danmaku content on the right side of the TV screen. The *voice danmaku* design involved danmaku into audio, allowing users to hear previous viewers' comments while watching TV, as referenced in [41]. The concept of *danmaku bubbles* was inspired by the idea of enclosing text content in bubbles that float through the VR space, appearing or disappearing gradually. The *virtual agents with bubbles* design was influenced by the virtual student concept from [20], mapping danmaku to virtual agents displayed in text bubbles. Finally, the *virtual agents with voice* design combined synthesized audio danmaku with virtual agents, enabling the agents to vocalize the content, inspired by the virtual audiences from [11]. Following these discussions, we voted for the *virtual agent with voice* design from the five best candidates (Figure 1).



Figure 1: Five design candidates following design discussions.

4 Implementation

Following the formative study, we designed and implemented a Danmaku Avatar system in a virtual living room environment using Unity (Figure 2). The danmaku avatars were equipped with both verbal and non-verbal behaviors such as gestures, head movements, and facial expressions derived from danmaku comments. Below, we present a module-by-module discussion of the danmaku avatar, detailing its implementation from danmaku comments to the final avatar.

Danmaku Classification. First, each danmaku is assigned an emotional label based on Russell's circumplex emotional model (a valence-arousal framework) [31] through a sentiment classification model, thereby identifying its affective state. We conducted

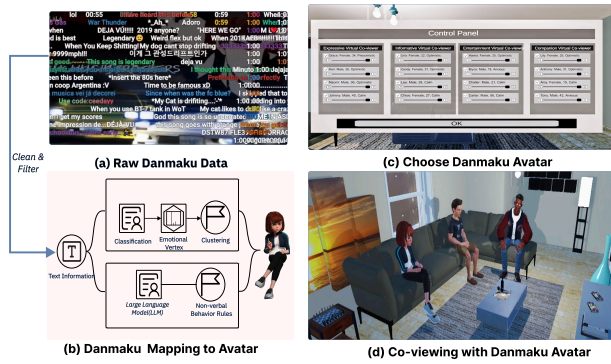


Figure 2: The design process of the Danmaku Avatar system involves four key stages: (a) raw danmaku data cleaning and process; (b) processing and clustering the comments to map them onto avatars, which enable real-time interactions supported by LLMs and non-verbal behavior rules; (c) danmaku avatar choose panel; (d) experiencing video watching alongside danmaku avatars within a VR environment.

sentiment classification on the danmaku using the DaLC sentiment classification dataset, published by Niu et al. [27]. Each danmaku comment was labeled with at least one multidimensional emotion descriptor based on the revised Russell’s circumplex model [31]. We then fine-tuned and evaluated several models, including BERT [7] and ERNIE 3.0 [33]. Finally, we utilized BERT models to further fine-tune the danmaku data from the DaLC dataset. Drawing on the danmaku content classification criteria proposed by [4] and [7], each danmaku is then classified into one of five types: entertainment, companionship, a sense of belonging, information seeking, and others. Specifically, we recruited three participants with research backgrounds in danmaku from a local university to organize and label the danmaku based on predefined classification criteria. Any inconsistencies or ambiguities encountered during the process were resolved through group discussions. Each annotator independently labeled 1,281 danmaku comments. Then, we selected BERT for further danmaku classification due to its superior performance.

Danmaku Clustering. To offer a more personalized viewing experience, we clustered the danmaku comments based on features such as emotional, content, and temporal attributes (Figure 2 (b)). We examined two temporal attributes—real-world time and video playback time—as both are crucial for analyzing danmaku distribution [15]. This clustering allowed comments with similar characteristics to be represented by the same danmaku avatar, symbolizing a group of viewers with shared interests. Consequently, users could choose one or more danmaku avatars to accompany them while watching a video, based on the avatars’ focus of interest, emotional traits, and visual design (Figure 2 (c)).

Verbal Behaviors. For each danmaku comment, we employ Tencent Cloud’s Text-to-Speech (TTS) technology⁴ using sentiment classification results to generate emotionally expressive speech with varying timbres, intonations, and tones.

Non-verbal Behaviors. We draw upon the methods outlined in [12] to design a set of non-verbal behavior rules for danmaku avatars in VR scenarios. These danmaku avatars perform actions such as gestures, head movements, and facial expressions according to these rules, which are tailored to align with the emotional context and the intended interaction goals. The rules consist of parameters that define various non-verbal behaviors, specifying particular actions and the likelihood of their occurrence. We designed a non-verbal behavior library for danmaku avatars of different genders and matched corresponding actions for each emotion based on the arousal and valence dimensions [17, 23], enhancing the naturalness of emotional expression and interaction. For instance, if a danmaku avatar needs to express “pleasure”, the rules indicate a 20% chance that the danmaku avatar will act as “arms bent, index finger extended for a light tap” while speaking. These non-verbal behaviors are carefully selected by the system designers from the MPI Emotional Body Expressions Database [35], Mixamo⁵, and Actorcore⁶ motion libraries.

Real-time Feedback We developed dialogue agents powered by a large language model (LLM) to provide real-time interactive support for the avatars [36]. The danmaku avatar actively voiced danmaku comments and displayed corresponding gestures as the video progressed. Users could interact with the avatars, prompting them to respond based on both the ongoing plot and their previous danmaku comments, thereby enhancing the co-viewing experience (Figure 2 (d)).

5 User Study

To address RQ3 and RQ4, we designed and conducted a comparative experiment to evaluate the user experience of danmaku avatars in asynchronous co-viewing scenarios.

5.1 Participants and Experiment Setup

We recruited 36 participants, 20 females and 16 males, with an average age of 23.15 (SD = 1.92) (N1-N36), which interested in the research through word-of-mouth and local university chat groups. The participants were randomly assigned into two groups of 18: one group watched the video alone with text format danmaku overlapping on the screen in a VR environment (VR-A group), and the other group watched the video with danmaku avatars (VR-DA group). Each participant was guided to a prepared room with HTC VIVE headsets, controllers, two base stations, and a BrainLink⁷ electroencephalography (EEG) headset. A technical assistant monitored and recorded the sessions from a separate room using a VR-capable laptop and OBS Studio⁸.

5.2 Measurement

We focused on measuring three key aspects: 1) the viewing experience in VR, including participants’ engagement and recall ability; 2) social presence, which measures the sense of social awareness while watching VR video with others [13]; and 3) the user experience of social interaction, measures participants’ behaviors and

⁴<https://www.tencentcloud.com/products/tts>

⁵<https://www.mixamo.com/>

⁶<https://actorcore.reallusion.com/3d-motion>

⁷<https://o.macrotellect.com/>

⁸<https://obsproject.com/>

feelings during interactions with the danmaku avatar, including the frequency and quality of discussing the plot and sharing emotions.

To compare the engagement levels between the two groups, we followed the methodology outlined by [2, 41], we reveal the engagement levels of users during the viewing process utilizing electroencephalography (EEG) signals, specifically from the α (7-13 Hz), β (13-30 Hz), and θ (4-7 Hz) frequency bands. Engagement (E) was calculated using the formula $E = \beta / (\alpha + \theta)$ to track changes in engagement during the video viewing. The engagement value was derived by calculating the average of the α , β , and θ frequencies. During the experiment, we employed three relevant subscales from the Social Presence Questionnaire (SPQ) [13], using a 5-point Likert scale, to assess social presence: Co-presence (CP), Perceived Message Understanding (PMU), and Perceived Affective Understanding (PAU). Finally, participants need to participate in a semi-structured interview.

5.3 Procedure

Upon arrival at the experimental room, participants were given a brief introduction and asked to sign a consent form. The experimenter then explained the research process and experimental procedure and guided participants in wearing the *HTC VIVE* headset and *BrainLink* EEG device, as well as using the controllers. Participants were subsequently given 5 minutes to acclimate to the VR environment by watching a short video. Participants in the VR-DA group were asked to select danmaku avatars to co-view with and were instructed on how to interact with them. During the experimental phase, participants in both the VR-DA and VR-A groups watched the same 20-minute video. At the end of the experiment, each participant completed the SPQ questionnaire followed by an approximately 15-minute interview.

5.4 Results

We employed both qualitative and quantitative methods to assess the effectiveness of danmaku avatars, focusing on engagement, social presence, and insights gathered from semi-structured interviews.

Engagement Analysis. The mean engagement score for the VR-DA group was 0.253 during the experiment, compared to 0.228 for the VR-A group, indicating a higher average for the VR-DA group. However, a t-test revealed a p-value of 0.502, suggesting no statistically significant difference between the two groups.

To observe how engagement levels fluctuated across the two groups, we divided the EEG data during the 20-min into five intervals, calculating the average engagement value for each four-minute interval based on video playback time, as shown in Figure 3. The results showed that engagement in the VR-A groups remained relatively stable, while the VR-DA group started with lower engagement that gradually increased over time. This trend may be attributed to the participant's familiarity with the viewing modes in the VR-A group, allowing for quicker adaptation. In contrast, participants in the VR-DA group needed time to acclimate to the novel asynchronous viewing model. Once adopted, their overall engagement levels surpassed those of the VR-A group. This trend warrants further exploration in future research, particularly considering the time required for participants to acclimate to the new environment.

Social Presence Analysis. In terms of social presence (Figure 4), the VR-DA group scored higher on all three subscales of the SPQ compared to the VR-A group: $VR-DA_{CP} = 4.533$, $VR-DA_{PMU} = 4.25$, $VR-DA_{PAU} = 0.373$; $VR-A_{CP} = 3.6$, $VR-A_{PMU} = 3.6$, $VR-A_{PAU} = 0.267$. One-way analysis of variance (ANOVA) showed that the VR-DA group significantly outperformed the VR-A group in the CP and PAU subscales ($p_{CP} < 0.05$, $p_{PMU} = 0.511$, $p_{PAU} < 0.05$), suggesting that danmaku avatars enhance social presence by promoting a sense of co-presence and improving emotional connection among participants.

User Feedback. Based on the interviews, we found that the higher standard deviation in PMU might be attributed to some participants' discomfort with using voice for danmaku. They felt that certain humor and nuances in danmaku were better conveyed through text. For instance, one participant noted, "The emotional expression in danmaku is more direct, and sometimes, the danmaku avatar made me feel a bit uncomfortable" (N10). Conversely, participants also reported experiencing enjoyment when the danmaku avatar expressed similar opinions beside them. As one stated, "When they say something that aligns with my thoughts right next to me, the experience is more engaging in moments of resonance" (N21).

All participants interacted with the danmaku avatar at least three times, with nearly all expressing that its presence increased their motivation to communicate. One participant shared, "I find it difficult to give a perfect score, but the voice responses are definitely more engaging than text-based danmaku" (N7). Another remarked, "There are times when I just want to vent without expecting any response. But when the avatar does respond, it's a pleasant surprise" (N32). However, excessive participation from danmaku avatars could sometimes detract from the experience, leading to distractions or interfering with the video's dialogue. As one participant described, "The danmaku avatar and the characters in the video spoke simultaneously, making it difficult to hear both clearly" (N4).

6 Discussion, Limitation, and Future Work

The primary aim of this study is to explore the transformation of danmaku—a text-based co-viewing experience on 2D screens—into a VR environment, enabling a co-viewing experience in VR and examining the resulting changes in user viewing behavior. Our

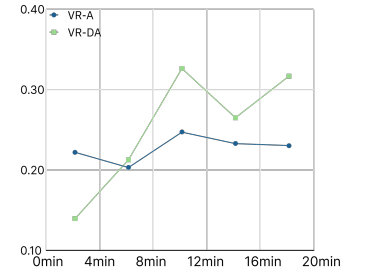


Figure 3: Changes in engagement over time during watching.

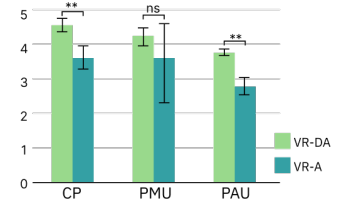


Figure 4: SPQ result compare. Error bars indicate standard deviations.(ns: $p > .1$, *: $p < .05$, **: $p < .01$)

findings suggest that while danmaku avatars did not significantly impact user engagement, they did enhance social presence and the desire for social interaction.

By mapping danmaku with similar characteristics to specific danmaku avatars—representing the perspectives of particular audience segments—users can select danmaku avatars whose viewpoints align with their own, thereby fostering emotional resonance and deepening immersion. However, designers must carefully balance the viewing experience with social interaction. A more optimal solution would involve dynamically adjusting danmaku avatar speech and behavior based on video content, preventing interruptions at key moments, and ensuring a seamless, uninterrupted viewing experience. Furthermore, participants were generally satisfied with the real-time responses from danmaku avatars, suggesting that LLM-based interactions effectively address the delays seen in traditional asynchronous co-viewing. However, while using plot content and danmaku as a knowledge base allows danmaku avatars to generate relevant responses, this approach falls short of meeting users' broader conversational needs. Danmaku avatars may make mistakes in these areas, as they rely on a generic LLM without specific background information. Therefore, it is essential to design danmaku avatars with more comprehensive social interaction capabilities, supported by detailed personal profiles, to accommodate the varied preferences of users.

Although this study provides an initial approach to transforming danmaku into danmaku avatars and offers preliminary insights into its impact on the user viewing experience, it has several limitations that warrant further exploration. First, in the current work, danmaku is clustered and mapped onto virtual human avatars, but future research could investigate more innovative representations of danmaku in VR environments to create new interactive and visual experiences. Second, based on user feedback, some danmaku are not well-suited for voice representation and are better expressed in text form. Future research will explore methods to process these danmaku to ensure their original meaning is preserved in voice-based expressions. Moreover, based on the findings from engagement research, we believe that the current experimental design has certain limitations. Future studies should not only examine the impact of different types of video content on engagement but also further investigate the process by which users acclimate to danmaku avatars. Moreover, users' cognitive and emotional responses to danmaku avatars may evolve over time, assessing their adaptation process an important factor in evaluating changes in engagement.

7 Conclusion

In summary, this work contributes valuable insights into the study of asynchronous viewing experiences in VR, particularly in the fields of human-computer interaction and user experience enhancement within virtual environments. It demonstrates how technology can be integrated with social behaviors to redefine how people co-view and interact in digital spaces, thus transforming the social aspects of content consumption. The exploration of the intersection between technology and social behavior could open up new avenues for applications in virtual social interactions and virtual reality.

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